

## Article

# Temperature Data Correction of Fast Updating Assimilation System Based on Machine Learning Algorithms

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## Abstract

In order to improve the accuracy of near-ground temperature forecasting under winter rain, snow, and freezing weather conditions, three machine learning algorithms, namely neural network, random forest, and support vector machine, were used to train various ground and air elements in the rapidly updated assimilation model of Zhejiang Province based on multi-source observation data, reducing the error of temperature in the model field and forming hourly and 3 km horizontal resolution ground and air temperature datasets for two rainy, snowy, and frozen weather processes. After calibration using the backpropagation neural network algorithm, random forest algorithm, and support vector machine algorithm, the MAE of the simulated field temperature forecast decreased from 1.29 °C to 0.937 °C, 1.01 °C, and 0.988 °C, respectively. The backpropagation neural networks and support vector machine algorithms perform well, but support vector machine algorithms have relatively short computation times. Using 10 feature points for training achieves optimal performance; more points may not necessarily lead to better calibration results. Adding actual data at the initial time of the target point significantly improved the correction effect, and the improvement effect was even better when the forecast lead time was less than 10. The correction effect of the prediction field shows that when the forecast lead time is between 15 h and 24 h, it becomes unstable over time. The mean prediction accuracy of whether the temperature exceeds the 0 °C temperature threshold at 24 forecast moments before calibration is 0.928. After correction, it has been increased to 0.956.

**Keywords:** temperature; machine learning algorithms; forecast; correction



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## 1. Introduction

Accurate temperature forecasting plays a critical role in mitigating ice-related disasters on transmission towers and ensuring the stability of power infrastructure [1,2]. Lu et al. [3] pointed out that ice accumulation on power lines and towers, driven by subfreezing temperatures and precipitation, can lead to structural failures, widespread power outages, and significant economic losses. Precise temperature predictions enable utilities to implement preventive measures, such as de-icing operations or load adjustments, minimizing disruption risks. Beyond power systems, reliable temperature forecasts are vital for agriculture, transportation, and emergency planning. Farmers depend on them to protect crops from frost, while the logistics and aviation sectors optimize operations to avoid weather-induced delays [4]. Thus, advanced meteorological modeling and real-time monitoring are essential for safeguarding both critical infrastructure and daily socioeconomic activities [5].

In the past few decades, with the rapid development of high-performance computers, numerical models, observation systems, and other technologies, the accuracy of numerical model forecasting has been greatly improved. Mesoscale models, such as the rapidly updated assimilation model of Zhejiang Province (ZJ3km model field), that incorporate diverse radar, satellite, and observational data, offer instrumental support for precise temperature forecasts [6]. However, factors such as physical framework, initial field, and boundary conditions can still affect the accuracy of numerical model prediction [7,8]. Using relevant equipment for on-site temperature measurement can obtain more accurate temperature data, but it is difficult to obtain predicted values and the cost is too high.

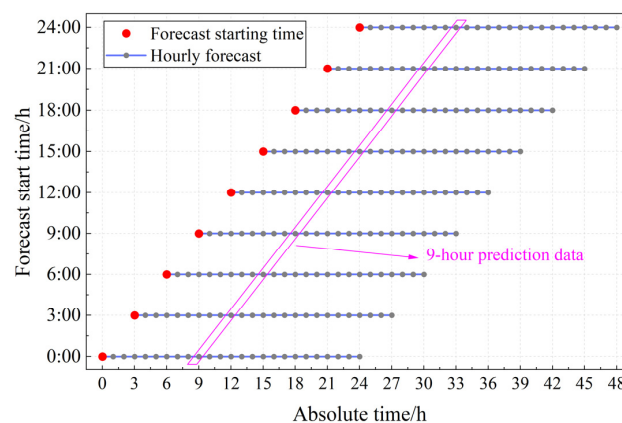
Therefore, the correction of temperature forecast products based on machine learning methods has become a research hotspot in the field of meteorology in recent years [9–16]. By combining the output of numerical weather forecast models with machine learning algorithms, the accuracy and reliability of forecast products can be improved. Whether the temperature is below zero degrees has a significant impact on natural disasters. For example, when the temperature is below zero degrees, there may be ice cover on outdoor transmission lines. Therefore, in the process of calibrating temperature data, attention should also be paid to whether the temperature exceeds the threshold of 0 °C. He et al. [17] used machine learning methods such as random forest and support vector machine to spatially interpolate the temperature in China's land areas and achieved significant results. Fan [18] considered the high spatial correlation of temperature forecast data and integrated the spatiotemporal feature information of temperature forecast errors from Attention-Dense U-net and Simple Video Prediction models to construct a multi-factor spatiotemporal temperature forecast correction model. Compared to the temperature forecast of the global forecast system before correction, the error after correction has decreased by 43%. Zhang et al. [19] used three machine learning methods, linear regression, LSTM-FCN, and LightGBM, to calibrate temperature forecasts for an operable high-resolution model GRAPES-3km. RMSE decreased by 33%, 32%, and 40%, MAE decreased by 33%, 34%, and 41%, and  $R^2$  increased by 21.4%, 21.5%, and 25.2%, respectively. LightGBM performed the best with a forecast accuracy of over 84%. Meng et al. [20] proposed a machine learning-based framework for spatial downscaling and bias correction of high-resolution temperature forecasts. This framework can downscale low resolution forecast data and correct biases to generate high-resolution temperature forecasts. Experimental results have shown that this framework has superior performance and potential in downscaling and correction of meteorological data. Soukayna et al. [21] introduced an adaptive bias correction method that combines state-of-the-art dynamic forecasting with observations using machine learning. When applied to the leading sub seasonal model of the European Centre for Medium Range Weather Forecasts, temperature forecasting skills improved by 60–90%. Liu et al. [22] developed a direct automated machine learning method for post-processing the raw forecasts of WRF models. This method was implemented and evaluated on post-processing forecasts from 13 stations in northern Xinjiang, where the average RMSE value decreased from 3.24 °C to 2.34 °C, a decrease of 28%.

To improve the accuracy of temperature forecasting in rainy, snowy, and freezing weather conditions, three machine learning algorithms, namely neural network, random forest, and support vector machine, were used to train various ground and air elements in the ZJ3km model field based on multi-source observation data, reducing the error of temperature in the model field and forming hourly and 3 km horizontal resolution ground and air temperature datasets for two rainy, snowy, and frozen weather processes. This study will offer enhanced precision in forecasting temperature data to assist power icing warnings and ensure operational safety during rainy, snowy, and icy weather conditions.

## 2. Data Measurement

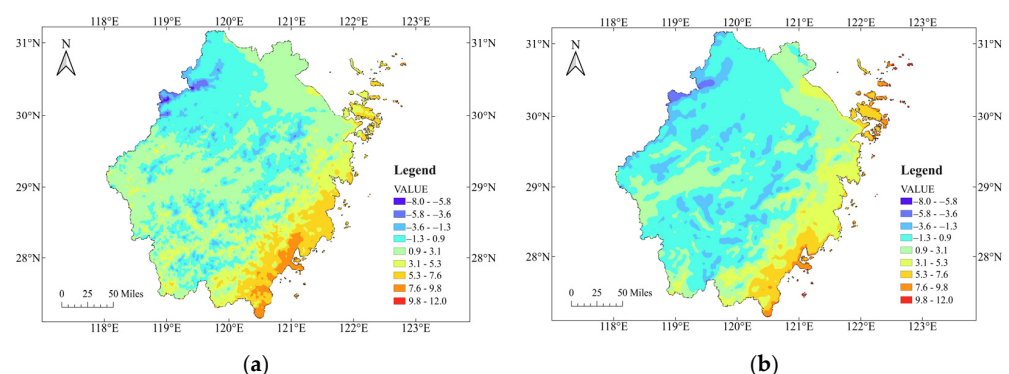
The dataset comprises simulated field temperature data from the rapidly updated assimilation model of Zhejiang Province (ZJ3km model field) alongside actual field data. The two categories of temperature data pertain to the periods of 29 November 2022 to 3 December 2022 and 21 February 2024 to 27 February 2024 in Zhejiang Province, associated with two instances of precipitation, snowfall, and freezing conditions. The authentic field data is derived from meteorological observation data and atmospheric temperature profiles obtained by microwave radiometers and is hourly data. The spatial sampling density of both types of temperature data is  $3 \text{ km} \times 3 \text{ km}$ , and there are a total of 37,944 sampling points.

The simulated field temperature data follows a 3-hourly initialization cycle, as schematically shown in Figure 1. Specifically, a new forecast is launched at 0:00, 3:00, 6:00, ..., 24:00 of each day, denoted by the red dots in Figure 1. Each forecast run generates 24 consecutive hourly temperature predictions, spanning from the initialization time to 23 h ahead, represented by the gray dots and blue lines. Therefore, the simulated field data can be classified as 0 h, 1 h, 2 h, ..., 24 h prediction data, and the 9 h prediction data are shown in Figure 1. The 0 h prediction data is also known as initial field prediction data.



**Figure 1.** The dataset of the simulated field for one day.

For predicted data at any given time, there are corresponding actual field data at the same time. Figure 2 shows the actual field data and simulation field data from 1 December 2022, at 00:00:00 in Zhejiang Province, China. The distribution of the two temperature datasets is notably analogous, with regions of reduced dimensions exhibiting elevated temperatures and regions of increased dimensions displaying diminished temperatures. Nonetheless, a substantial disparity in temperature values persists between the two temperature data, and the simulated field temperature is relatively low.

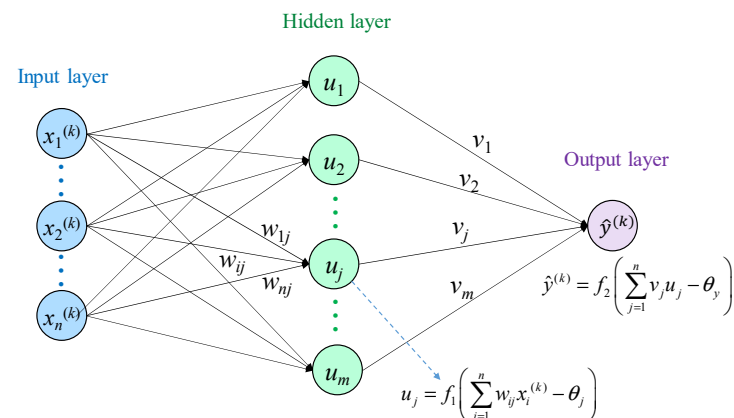


**Figure 2.** Temperature on 1 December 2022 at 00:00:00: (a) Actual field data. (b) Simulation field data.

### 3. Machine Learning Algorithm

#### 3.1. GA-BP Algorithm

The backpropagation neural network (BPNN) [23,24] is a fundamental supervised learning algorithm for multilayer feedforward neural networks, which has been widely applied in nonlinear function approximation, regression, and classification tasks due to its powerful ability to model complex data relationships. As illustrated in Figure 3, the network adopts a three-layer fully connected structure consisting of an input layer for receiving raw feature vectors, a hidden layer for nonlinear feature transformation, and an output layer for generating the final prediction.



**Figure 3.** Schematic diagram of backpropagation neural network.

In the forward propagation phase, input signals are transmitted layer-by-layer through weighted synaptic connections. For the  $j$ -th hidden neuron, the net input is computed as the weighted sum of input features minus a bias term  $\theta_j$ , followed by an activation function  $f_1(\cdot)$  to produce the hidden output:

$$u_j = f_1 \left( \sum_{i=1}^n w_{ij} x_i^{(k)} - \theta_j \right) \quad (1)$$

where  $w_{ij}$  denotes the connection weight between the  $i$ -th input neuron and the  $j$ -th hidden neuron.

The hidden layer outputs are then propagated to the output layer, where the final prediction can be represented as follows:

$$\hat{y}^{(k)} = f_2 \left( \sum_{j=1}^m v_j u_j - \theta_y \right) \quad (2)$$

where  $v_j$  denotes the weight from the  $j$ -th hidden neuron to the output neuron,  $\theta_y$  the output layer bias, and  $f_2(\cdot)$  the output activation function tailored to the specific task.

The core of the BP algorithm lies in the backward propagation of prediction errors. After computing the discrepancy between the network output and the ground-truth label, the error is propagated backward from the output layer to the input layer, and all connection weights and biases are updated iteratively using the gradient descent method to minimize a predefined loss function. This iterative forward-backward process continues until the network converges to a predefined error threshold or reaches the maximum number of training epochs, enabling the network to learn the implicit nonlinear relationships embedded within the training dataset.

### 3.2. Random Forest Algorithm

Random Forest (RF) [25] is a classic ensemble learning algorithm based on decision trees, and its core logic can be intuitively understood through Figure 4. The schematic displays a single decision tree's structure: internal nodes represent feature split conditions (e.g.,  $x_1 < s_1$ ,  $x_4 < s_4$  and leaf nodes ( $R_1$ – $R_6$ ) are prediction regions. RF overcomes the limitations of single decision trees, such as overfitting and high variance, by introducing dual randomness and ensemble mechanisms.

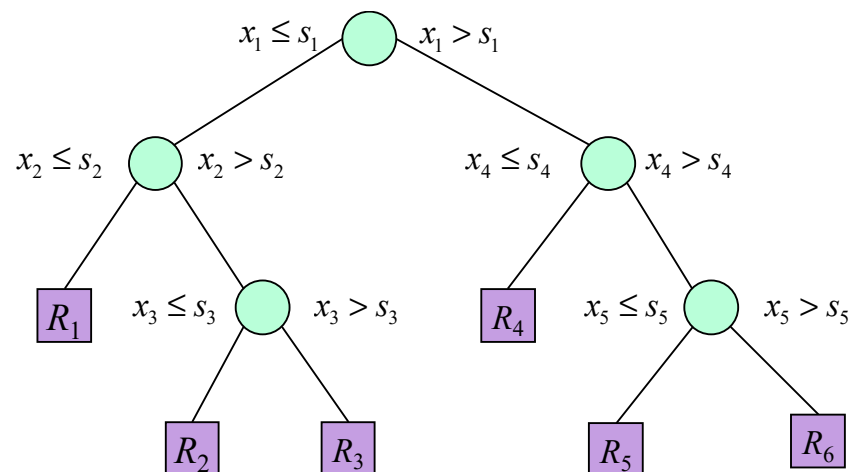


Figure 4. Decision tree flowchart.

As shown in Figure 5, the training of RF consists of two core steps. First, bootstrap sampling is performed on the original dataset to generate  $K$  independent sub-datasets, each used to train a decision tree. For each node split of a tree, a random subset of features is selected, instead of using all features. The optimal split threshold is chosen based on impurity metrics. This random feature selection ensures diversity among individual trees, reducing the correlation between them. Each tree is grown to its maximum depth without pruning, and  $K$  such independent trees form the final random forest.

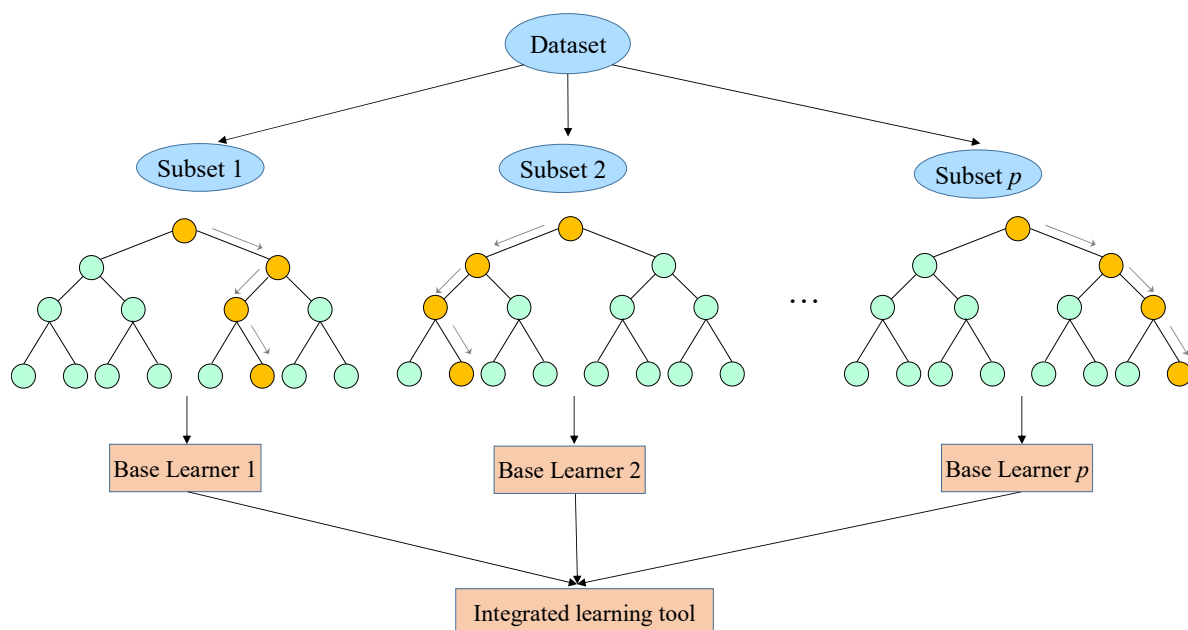
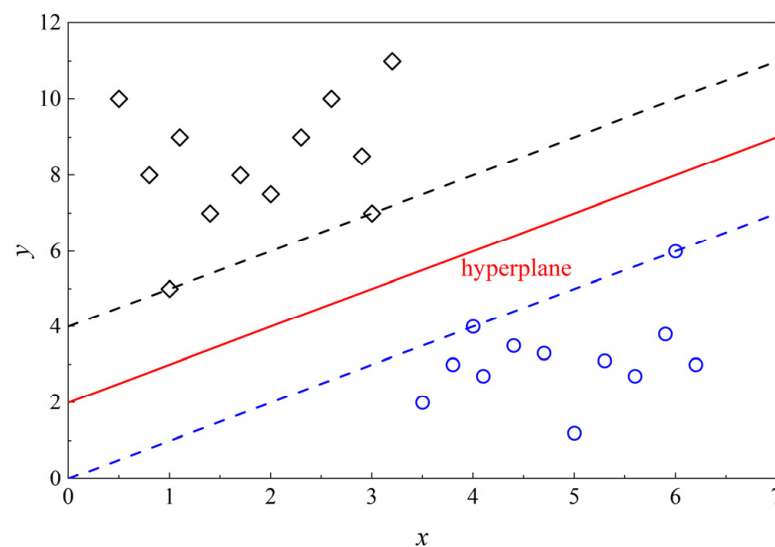


Figure 5. Random forest process diagram.

During inference, input samples traverse all trees to generate base predictions, aggregated via majority voting or mean aggregation. The schematic's leaf nodes exemplify single-tree decision logic, with ensemble averaging enhancing generalization. RF's strengths lie in dual-randomness-derived overfitting resistance, high accuracy, noise resilience, and built-in feature importance evaluation. Limitations include reduced interpretability, higher computational overhead, and subpar performance on strongly correlated features.

### 3.3. Support Vector Machine Algorithm

Support Vector Machine (SVM) [26–28] is a powerful supervised learning algorithm primarily designed for classification tasks and extended to regression and outlier detection. It aims to find an optimal hyperplane that maximizes the margin between different classes in the feature space. The hyperplane is determined by a subset of training samples called support vectors, which lie closest to the hyperplane and play a decisive role in defining the decision boundary, as shown in Figure 6.



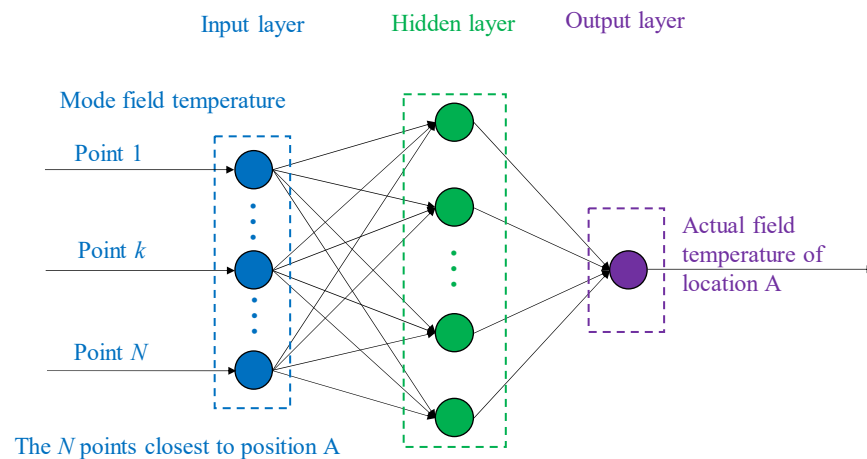
**Figure 6.** The hyperplane of the support vector machine.

For linearly separable data, SVM constructs the maximum-margin hyperplane by minimizing the norm of the weight vector subject to classification constraints. For non-linearly separable data, it employs kernel functions to map the original feature space into a higher-dimensional Hilbert space where linear separation is feasible. Common kernel functions include linear polynomial radial basis functions and sigmoid functions, each adapting to different data distribution characteristics.

SVM exhibits excellent generalization ability due to its maximum-margin principle, which effectively reduces the overfitting risk. It performs well with high-dimensional data even when the number of features exceeds the number of samples, making it widely applicable in fields such as pattern recognition, image processing, bioinformatics, and text classification.

### 3.4. Correction Method and Cross-Validation

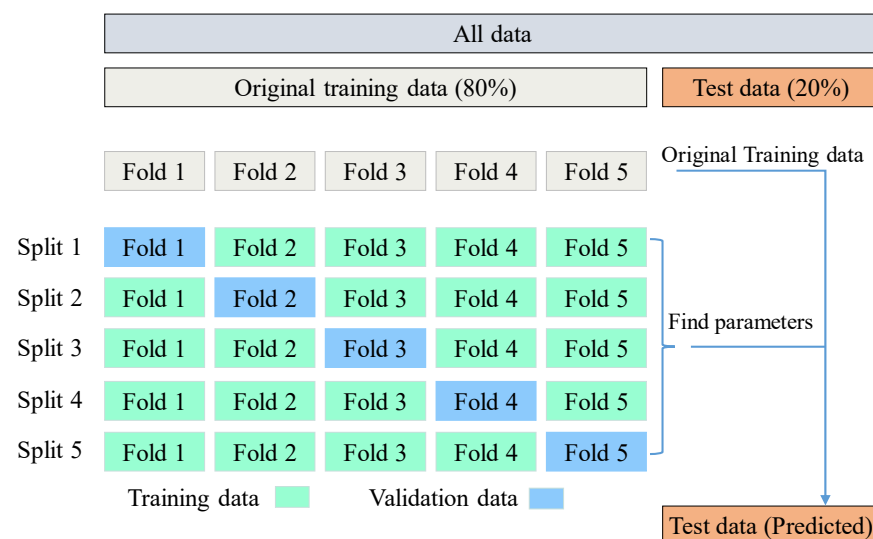
Amend the pattern field data in accordance with actual field data. In rectifying the temperature data for a specific place A, the temperature data from the N closest locations to that site is utilized. The input feature is represented by A in the model, whereas the actual temperature data for location A serves as the output label. The rectification process is illustrated in Figure 7, using the neural network algorithm as a reference.



**Figure 7.** Input features and output labels.

In the construction of machine learning models, appropriate sample partitioning and dependable performance evaluation are essential for the model's generalization capability. The input sample dataset is segmented into two components: training samples and test samples. The training samples are mostly utilized for model learning and training, whilst the test examples are employed to assess the generalization capacity of the taught machine learning model.

A total of 20% of the sample data is randomly designated as the test set, while the remaining 80% constitutes the original training set. To improve prediction accuracy and mitigate overfitting, cross-validation is frequently utilized to identify the ideal hyperparameters for machine learning. *K*-Fold Cross-Validation is extensively utilized in machine learning techniques. The fundamental principle entails evenly partitioning the original training set into *K* mutually exclusive subsets. In *K* rounds of iterative training and validation, one subset is designated as the validation set in each round, while the remaining  $K - 1$  subsets are amalgamated to constitute the training set, as depicted in Figure 8. The mean loss function values over the *K* validation sets are utilized to assess the efficacy of the hyperparameters. This method maximizes the use of limited data, hence decreasing the variance of evaluation outcomes and offering a more thorough depiction of the model's performance across various data subsets.



**Figure 8.** Process diagram for 5-fold cross validation.

The value of  $K$  is generally set to 5, 10, or 20 [29,30]. In this study, we employ a 5-fold cross-validation method to evaluate the hyperparameters of machine learning algorithms. The data partitioning and training process for 5-fold cross-validation are illustrated in Figure 8. Based on the optimized hyperparameters obtained from cross-validation, learning and training are performed on the original test set.

### 3.5. Error Evaluation Metrics

Three error metrics are used to evaluate the errors before and after correction, namely, the goodness-of-fit  $R^2$ , the mean absolute error (MAE), and the root mean squared error (RMSE), to assess the prediction performance of machine learning algorithms on the test set. The calculation formulas are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - Y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - Y_i| \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - Y_i)^2} \quad (5)$$

where  $n$  represents the sample size,  $y_i$  represents actual value,  $Y_i$  represents the predicted value,  $\bar{y}_i$  represents the mean value of actual values, and  $\bar{Y}_i$  indicates the average of predicted values. The range of the goodness of fit  $R^2$  is 0 to 1, and the closer it is to 1, the better the prediction performance. The closer the mean absolute error (MAE) and root mean square error (RMSE) are to 0, the better the prediction performance.

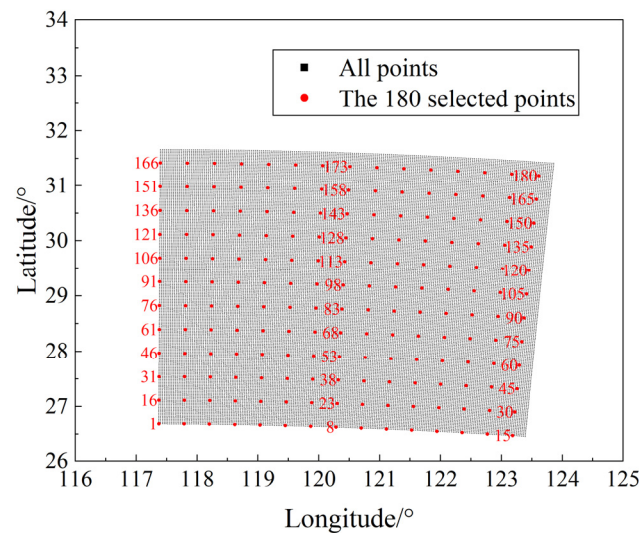
## 4. Results

### 4.1. Correction Results for Different Numbers of Feature Points

Liang et al. [31] pointed out that the terrain, landforms, and climatic conditions fluctuate significantly across several sites, necessitating distinct correction models. Independent training and prediction are necessary for each position to enhance accuracy. Given the extensive number of positions and the considerable time required, 180 positions are uniformly chosen for prediction. The mean of 180 positional error indicators is employed to assess predictive accuracy. Figure 9 illustrates 180 selected points together with their respective numerical identifiers. Enumerate the 180 selected points from west to east and from south to north, incrementing the numbering correspondingly, and the numbering of certain sites is shown in Figure 9.

When correcting the target point, first select the  $N$  nearest points to the point, take the simulated temperature data of the  $N$  points as the input feature, and use the actual field of the target point as the label. In addition, randomly select 20% of the data as the test set and 80% of the data as the training set.

The value of  $N$  may have a significant impact on the correction effect. Therefore, taking the support vector machine algorithm as an example, compare the correction effects when  $N = 1, 10$ , and  $20$ , as shown in Table 1. It can be observed that the best performance is achieved when using 10 feature points; more points do not necessarily lead to better correction results. When  $N$  was set to 10, after support vector machine algorithm correction, for the initial mode field temperature,  $A$  decreased by 0.05, MAE decreased by 27.4, and RMSE decreased by 25%. Therefore,  $N = 10$  corrects the temperature data.



**Figure 9.** The 180 selected points and their corresponding numbers.

**Table 1.** Temperature-correction effect (SVR).

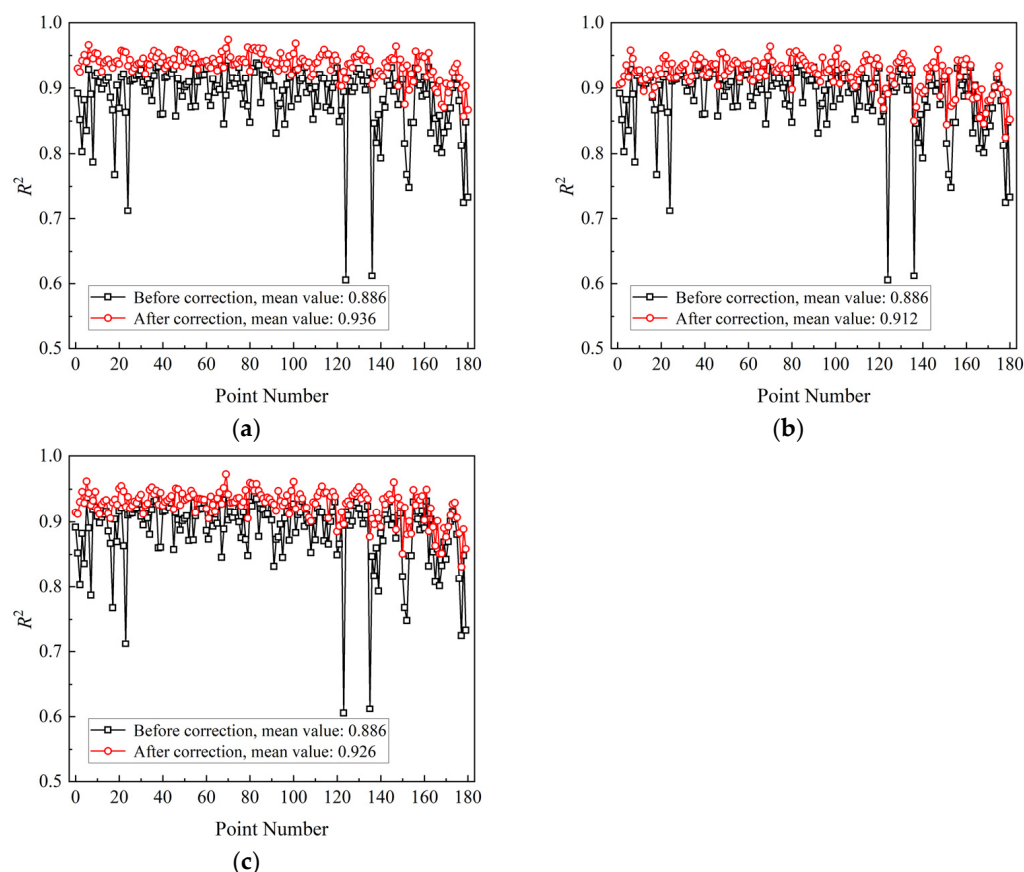
|                  |                   | $R^2$              | MAE ( $^{\circ}\text{C}$ )  | RMSE ( $^{\circ}\text{C}$ )  |
|------------------|-------------------|--------------------|-----------------------------|------------------------------|
| After correction | Before correction | 0.886              | 1.290                       | 1.663                        |
|                  | $N = 1$           | 0.915              | 1.056                       | 1.439                        |
|                  | $N = 10$          | 0.936              | 0.937                       | 1.248                        |
|                  | $N = 20$          | 0.935              | 0.930                       | 1.27                         |
|                  |                   | Increment of $R^2$ | Improvement rate of MAE (%) | Improvement rate of RMSE (%) |
| After correction | $N = 1$           | 0.029              | 18.1%                       | 13.5%                        |
|                  | $N = 10$          | 0.05               | 27.4%                       | 25.0%                        |
|                  | $N = 20$          | 0.049              | 27.9%                       | 23.6%                        |

#### 4.2. Correction Results of Different Algorithms

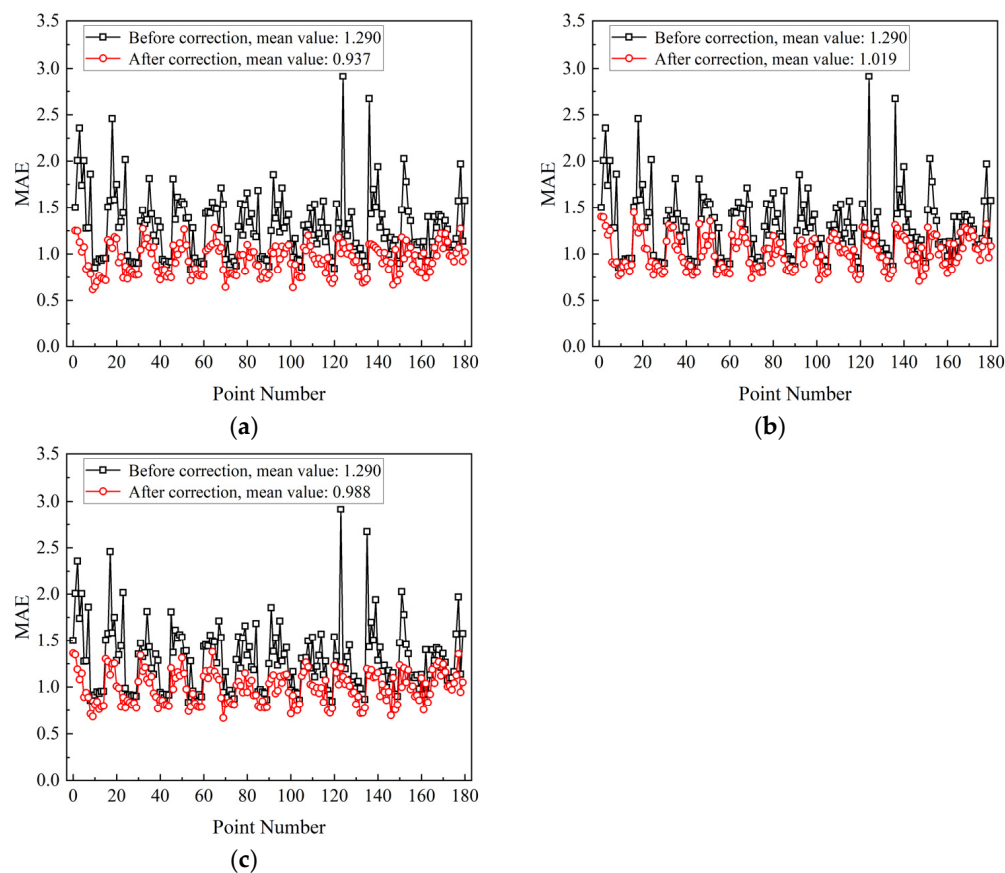
For 180 selected points, the correction results of the backpropagation neural network algorithm, random forest algorithm, and support vector machine algorithm are shown in Figures 10–12. The mean error index of the 180 selected points is shown in Table 2.

The goodness of fit  $R^2$  between the raw data and the actual data is 0.886. After correction by the neural network algorithm, random forest algorithm, and support vector machine algorithm,  $R^2$  reaches 0.936, 0.912, and 0.926, respectively. The MAE between the raw data and the actual data is 1.290. After correction by the backpropagation neural network algorithm, random forest algorithm, and support vector machine algorithm, MAE reaches 0.937, 1.019, and 0.988, respectively; the neural network algorithm performs well. In addition, it should be pointed out that although the error of the support vector machine algorithm is slightly larger than that of the neural network, its computation time is relatively short, with a computation time of about 2 h for 180 positions and about 13.5 h for the neural network. Wang and Qiao [32] also found that the support vector machine algorithm has significant advantages in reducing computation time. Therefore, the support vector machine is selected as the correction algorithm.

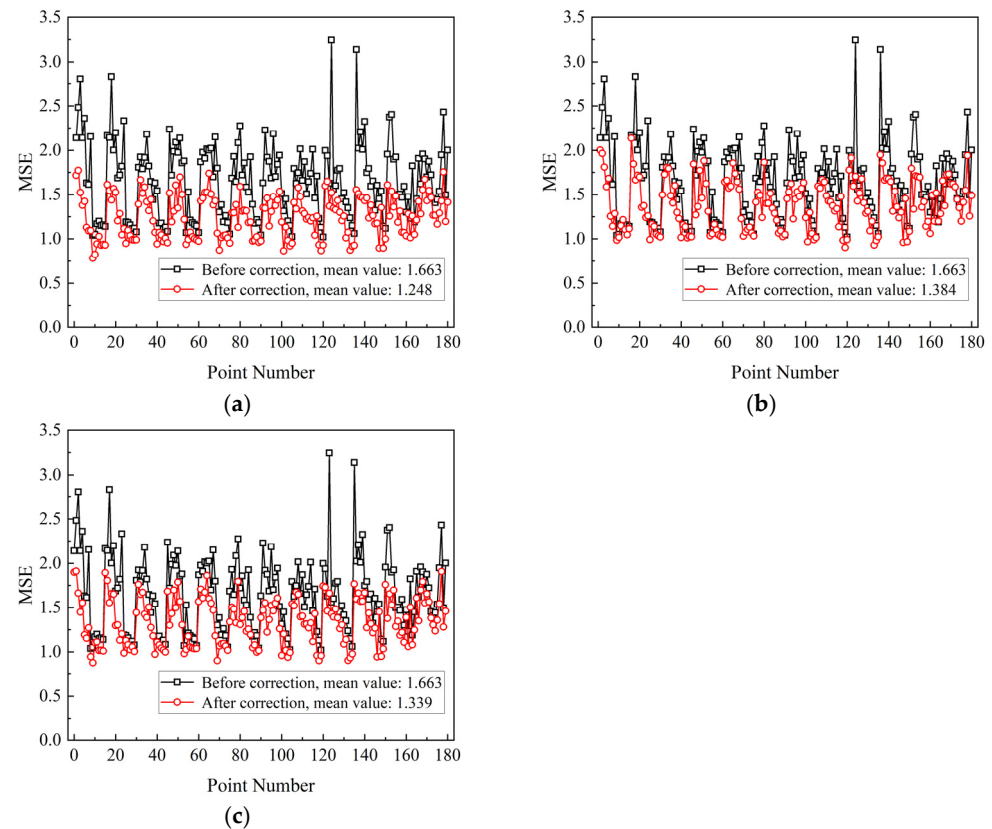
The comparison of the predicted values of the simulated field, both pre- and post-correction, to the actual values is illustrated in Figure 13 for points 76, 83, and 90. After correction by the SVR algorithm, the predicted data of the simulated field aligns more closely with the actual values.



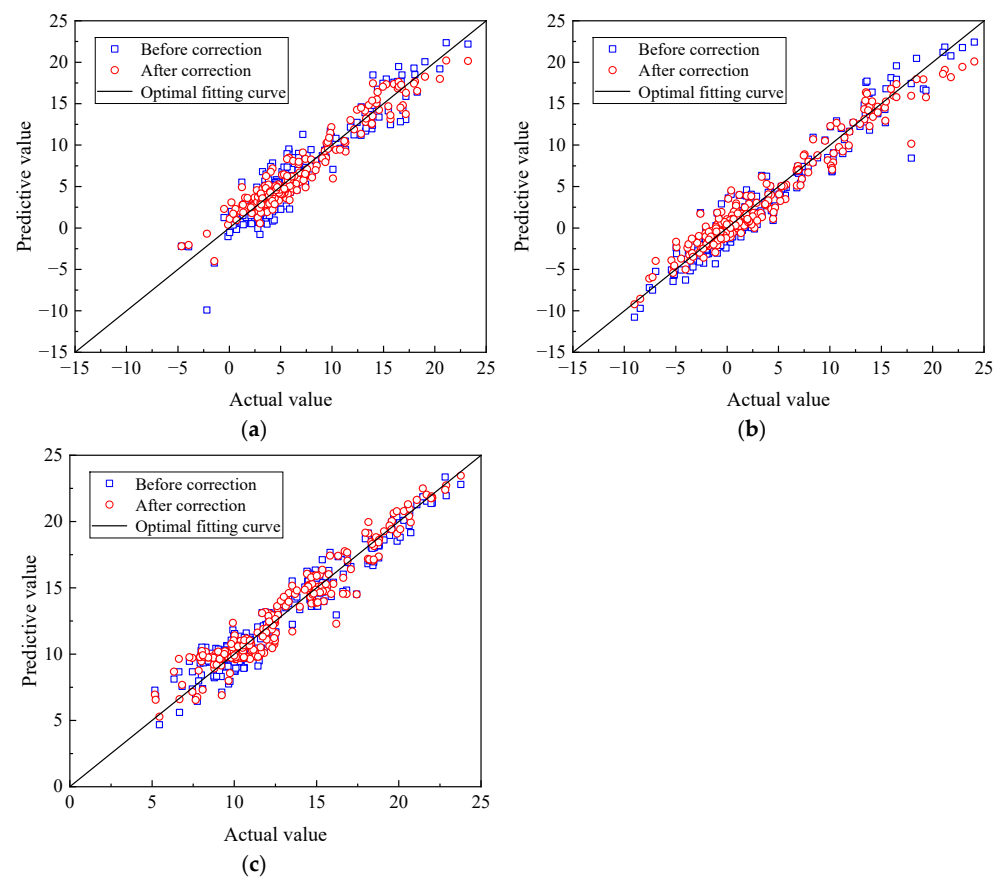
**Figure 10.** Mean  $R^2$  of 180 selected points: (a) BPNN. (b) RF. (c) SVR.



**Figure 11.** Mean MAE of 180 selected points: (a) BPNN. (b) RF. (c) SVR.



**Figure 12.** Mean MSE of 180 selected points: (a) BPN. (b) RF. (c) SVR.



**Figure 13.** Comparison of temperature data before and after correction (SVR): (a) Point 76. (b) Point 83. (c) Point 90.

**Table 2.** Temperature correction effect of three algorithms.

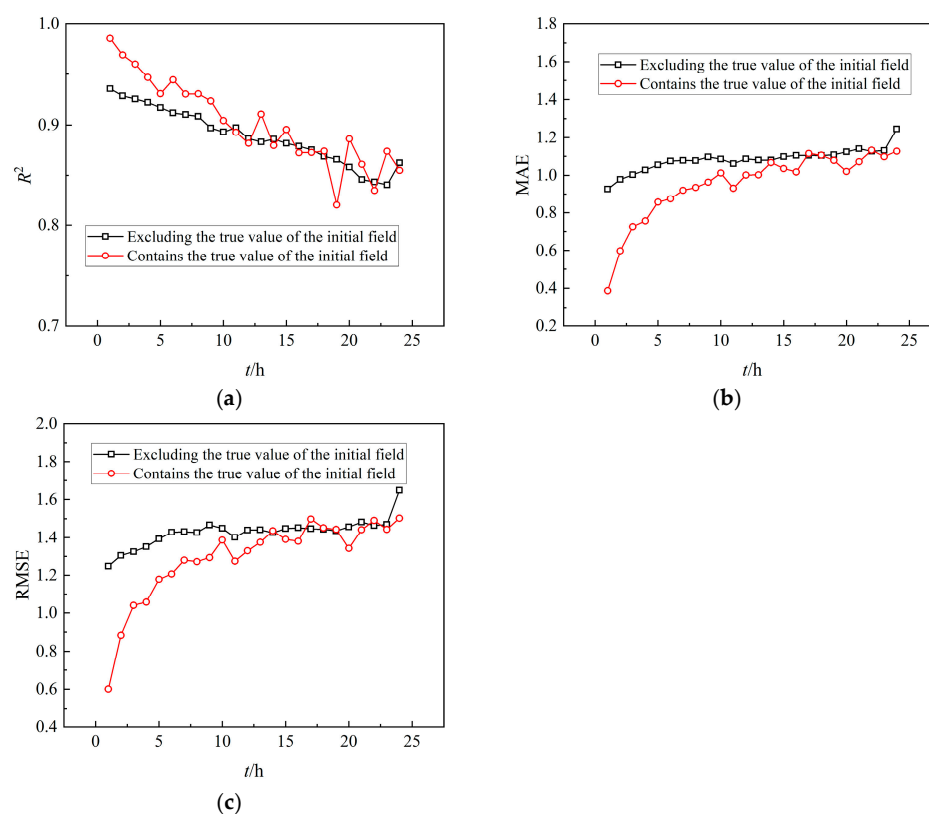
|                  |                   | $R^2$              | MAE (°C)                    | RMSE (°C)                    |
|------------------|-------------------|--------------------|-----------------------------|------------------------------|
| After correction | Before correction | 0.886              | 1.290                       | 1.663                        |
|                  | BPNN              | 0.936              | 0.937                       | 1.248                        |
|                  | RF                | 0.912              | 1.019                       | 1.384                        |
|                  | SVR               | 0.926              | 0.988                       | 1.339                        |
|                  |                   | Increment of $R^2$ | Improvement rate of MAE (%) | Improvement rate of RMSE (%) |
| After correction | BPNN              | 0.05               | 27.4%                       | 25.0%                        |
|                  | RF                | 0.026              | 21%                         | 16.8%                        |
|                  | SVR               | 0.04               | 23.4%                       | 19.5%                        |

## 5. Discussion

### 5.1. The Influence of Initial Field Prediction Data

To investigate the influence of the actual temperature data at the initial time of the selected point on the correction effect, the actual temperature at the initial time of the 180 selected points was used as the input feature for correction.

When the forecast lead time is 1–24 h, considering the actual data before and after the initial time of the 180 selected points, the correction effect of the forecast field data is shown in Figure 14. As the prediction duration increases, there is a trend of increasing prediction error, as found by Part et al. [15]. It can also be observed that the addition of actual data at the initial time of the 180 selected points significantly improves the correction effect, and the improvement effect of the forecast field is better the first 10 times. The correction effect of the forecast field shows instability over time from 15 h to 24 h.

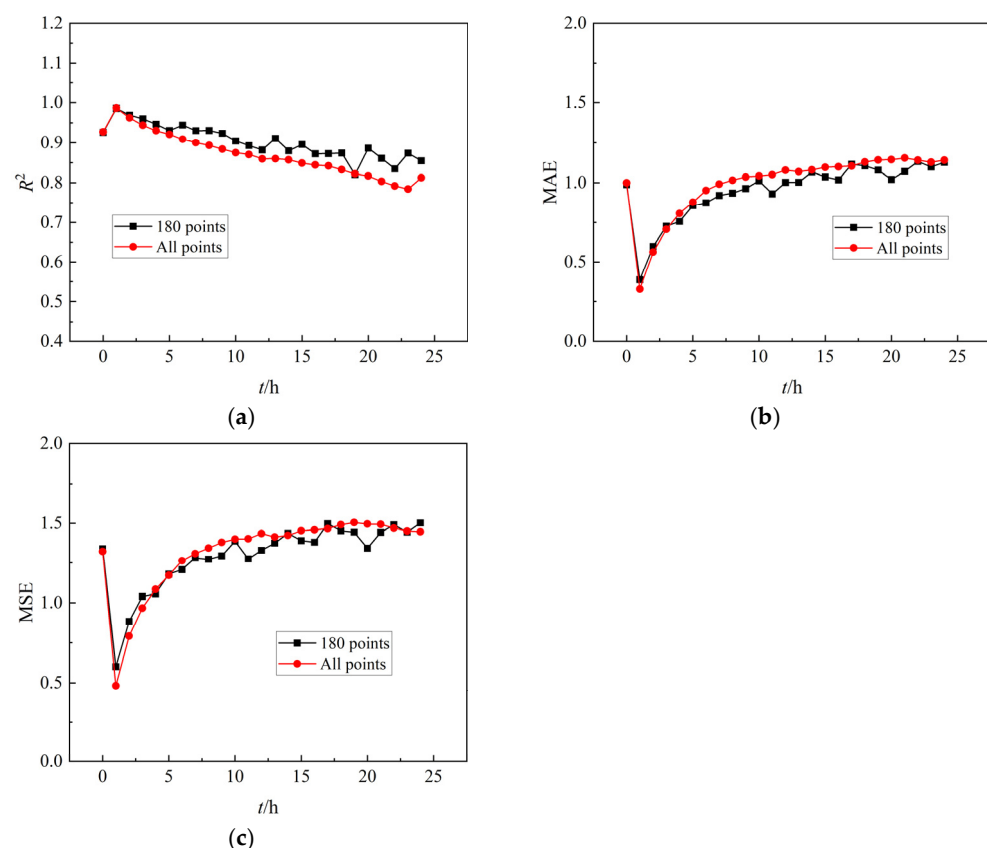


**Figure 14.** Correction effect before and after adding the selected point at the initial time: (a)  $R^2$ . (b) MAE. (c) RMSE.

Fan et al. [18] introduced an attention mechanism and dense connection module to construct a temperature-correction model, and the mean RMSE of temperature data decreased to 1.3, with an improvement ratio between 39.82% and 41.63%. The improvement ratio of SVR in this study was between 21.8% and 61.7%. However, the simulated field data used in this article is closer to the actual field, making temperature data correction more difficult.

### 5.2. Evaluation of Correction Effect of the Entire Field

For the global field of 37,944 points, the SVR algorithm was used to correct the temperature data of the simulated field from 00 to 24 h. The corrected effect is shown in Figure 15. It can be observed that the correction effect of selecting 180 points is very similar to that of all points, and the correction effect of 180 points is slightly greater. The uniformly selected 180 points are representative.

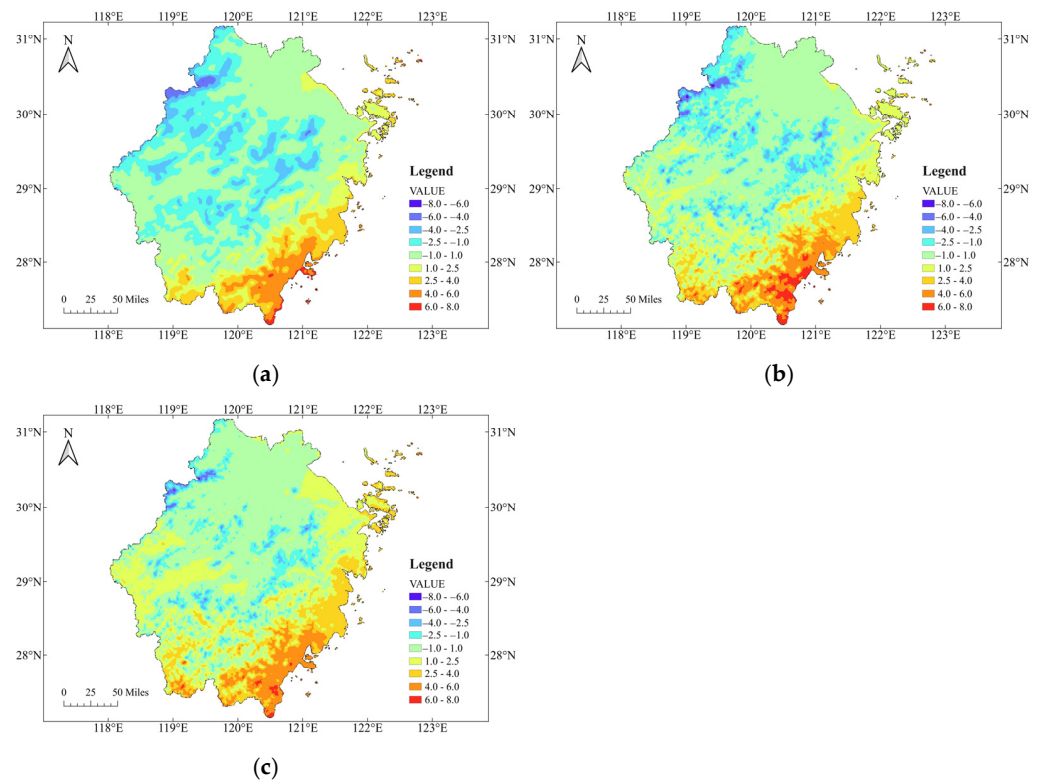


**Figure 15.** Comparison of correction effects between 180 points and all points: (a)  $R^2$ . (b) MAE. (c) RMSE.

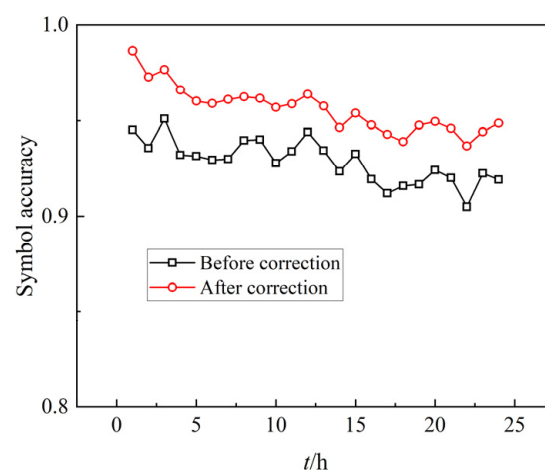
The comparison between the simulated field temperature data and the actual field temperature data at 21:00 on 22 February 2024 before and after correction is shown in Figure 16. It can be seen from Figure 16 that, compared to the real field temperature, the simulated field temperature before correction is lower, and the area below 0 °C is larger. After correction, the area of the simulated field below 0 °C has decreased and is closer to the actual field.

The temperature of 0 °C is crucial for whether ice has formed. The 0 °C temperature threshold is critical for ascertaining the occurrence of freezing. For example, Li et al. [33] pointed out that the transmission line may experience icing when the temperature is below 0 °C, which may lead to freezing disasters. This article categorizes temperatures below 0 °C as freezing and those above 0 °C as non-freezing. A binary classification test distinguishing

freezing from non-freezing conditions is performed on the temperature forecast field prior to and following rectification. The accuracy before and after correction is shown in Figure 17. The mean prediction accuracy of whether the temperature exceeds the 0 °C temperature threshold at 24 forecast moments before calibration is 0.928. After correction, it has been increased to 0.956.



**Figure 16.** Temperature on 22 February 2024 at 21:00:00: (a) Simulation field data before correction. (b) Simulation field data before correction. (c) Actual field data.



**Figure 17.** Accuracy of whether the temperature exceeds the 0 °C temperature threshold.

## 6. Conclusions

To improve the accuracy of temperature forecasting in rainy, snowy, and freezing weather conditions, this study mainly uses three machine learning algorithms to correct the temperature data of simulated fields in rainy, snowy, and freezing weather conditions. The conclusions can be summarized as follows:

1. When selecting the simulated field temperature closest to the target point as the input feature, the actual field temperature is used as the label. The best performance is achieved when using 10 feature points; more points do not necessarily lead to better correction results.
2. After calibration using the backpropagation neural network algorithm, random forest algorithm, and support vector machine algorithm, the MAE of the simulated field temperature forecast decreased from 1.29 °C to 0.937 °C, 1.019 °C, and 0.988 °C, respectively. The backpropagation neural networks and support vector machine algorithms perform well, but support vector machine algorithms have relatively short computation times.
3. Adding actual data at the initial time of the target point significantly improved the correction effect, and the improvement effect was even better when the forecast lead time was less than 10. The correction effect of the prediction field shows that when the forecast lead time is between 15 h and 24 h, it becomes unstable over time.
4. The mean prediction accuracy of whether the temperature exceeds the 0 °C temperature threshold at 24 forecast moments before calibration is 0.928. After correction, it increases to 0.956.
5. This work develops a revised model focused on winter precipitation, including rain, snow, and freezing events, which are primarily relevant to low-temperature meteorological phenomena in the complicated topography of the Zhejiang region. In the future, additional seasonal samples may be incorporated, and terrain and subsurface variables might be introduced to further augment the model's universality and its capacity to rectify long-term forecasts.

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